

# Probabilistic Machine Learning

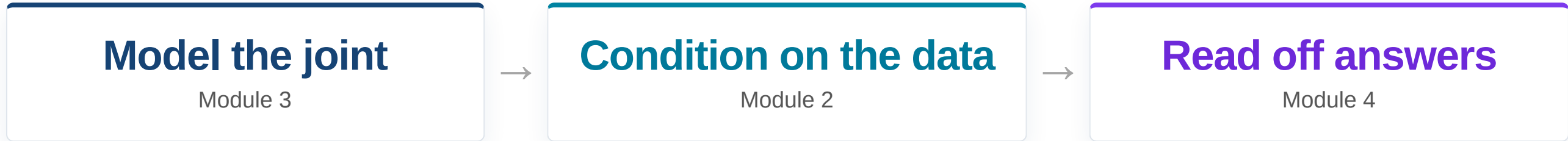
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## Lecture 02: Probability and Statistics

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# Where We Left Off

Lecture 01 ended on three moves, and promised that their vocabulary would be made precise here:



It also left a claim unpaid: that **epistemic** uncertainty is the kind more data removes. Module 4 turns that slogan into a rate.

# Module 1

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## A Language for Uncertainty

**Guiding Question:** *We need a language that can say "how strongly", not just "true or false". What must such a language satisfy — and who decides the numbers?*

# Example: Two Fruit Boxes

## Red box

2 apples, 6 oranges

## Blue box

3 apples, 9 oranges

We pick a box, then a fruit from it.

- If we pick the **red box** **40%** of the time, how often do we get an apple?
- We got an **orange** — which box did it probably come from?

# The Counts, in One Grid

Twenty picks, recorded as a table of counts — box across, fruit down:

|                  | $B = r$ | $B = b$ | row total |
|------------------|---------|---------|-----------|
| $F = a$ (apple)  | 2       | 3       | 5         |
| $F = o$ (orange) | 6       | 9       | 15        |
| column total     | 8       | 12      | 20        |

## Joint

a single cell over the total

$$P(B = r, F = a) = \frac{2}{20}$$

## Marginal

a row or column total over the total

$$P(F = a) = \frac{5}{20}$$

## Conditional

a cell over its own row or column

$$P(F = a \mid B = r) = \frac{2}{8}$$

# The Two Rules

## Sum rule

$$P(X) = \sum_Y P(X, Y)$$

Add along a row or column to remove a variable.

## Product rule

$$P(X, Y) = P(Y | X)P(X)$$

A cell is its column total times its share of that column.

Almost all of probability theory is these two rules applied repeatedly.

# Bayes' Theorem, in Advance

Combining the two rules and the symmetry  $P(X, Y) = P(Y, X)$ :

$$P(Y | X) = \frac{P(X | Y)P(Y)}{P(X)}, \quad P(X) = \sum_Y P(X | Y)P(Y)$$

**Which box did the orange come from?**

# Bayes' Theorem, in Advance

Combining the two rules and the symmetry  $P(X, Y) = P(Y, X)$ :

$$P(Y \mid X) = \frac{P(X \mid Y)P(Y)}{P(X)}, \quad P(X) = \sum_Y P(X \mid Y)P(Y)$$

Which box did the orange come from?

$$P(B = r \mid F = o) = \frac{P(F = o \mid B = r)P(B = r)}{P(F = o)} = \frac{\frac{6}{8} \cdot 0.4}{0.75 \cdot 0.4 + 0.75 \cdot 0.6} = 0.4$$

Here the fruit told us **nothing** about the box — both boxes have the same orange fraction. Evidence is only informative when the hypotheses predict different things.



# Takeaway

Two rules — **sum** to remove a variable, **product** to split a joint — generate joint, marginal, conditional probabilities and Bayes' theorem. The rest of this chapter makes them precise and extends them beyond counting.

# What Makes an Experiment "Random"?

**Definition (Random experiment).** A procedure that can — at least in theory — be repeated infinitely often, characterised by:

1. **Configuration** — a precisely specified system that can be reconstructed.
2. **Procedure** — an instruction for executing the experiment.
3. **Unpredictability** of the outcome.

**Coin flip:** a 1-Euro coin (configuration), placed on the thumb and snapped into the air (procedure), record heads or tails (outcome).

# Randomness as Missing Information

A random experiment is still causal. Its unpredictability reflects **missing information about the causal chain**, not an absence of causes.

Consequence: a random experiment can become a deterministic process once new information is available.

A coin flip filmed at 10,000 fps and measured precisely enough stops being random. What does that imply for a model that calls something "random"?

# Sample Space and Events

**Definition (Sample space, event space).** A set  $\Omega = \{\omega_1, \omega_2, \dots, \omega_n\}$  is the **sample space** of a random experiment if every outcome is associated with exactly one element  $\omega \in \Omega$ . The elements are called **outcomes**.

Let  $\Omega$  be finite. Every subset  $A \subseteq \Omega$  is an **event** the event  $A$  **occurs** iff the observed outcome  $\omega$  is a member of  $A$ . The set of all events, the power set  $\mathcal{P}(\Omega)$ , is the **event space**.

An outcome is a single result; an event is a *question you can ask* about the result.

For infinite  $\Omega$  — the drug dosage below — the event space is a suitable family of subsets rather than the full power set we return to this when continuous distributions appear.

# Important Event Types

$$\emptyset$$

impossible event

$$|A| = 1$$

elementary event

$$\Omega$$

certain event

$$A \subseteq B$$

$A$  is a sub-event of  $B$  — " $A$  entails  $B$ "

$$\overline{A} := \Omega \setminus A$$

complementary event

$$A \cap B = \emptyset$$

$A$  and  $B$  are incompatible

# Examples

## One coin flip

$$\Omega = \{H, T\}$$

$$\mathcal{P}(\Omega) = \{\emptyset, \{H\}, \{T\}, \{H, T\}\}$$

Event "heads":  $E = \{H\}$

## Two coin flips

$$\Omega = \{HH, HT, TH, TT\}$$

Event "at least one head":  $E = \{HH, HT, TH\}$

Four outcomes, sixteen possible events.

# The Event Space Grows Fast

**Three-sided die:**  $\Omega = \{1, 2, 3\}$ , so  $\mathcal{P}(\Omega)$  has  $2^3 = 8$  events, ordered by inclusion:

$\{1, 2, 3\}$                        $\{1, 2\}$     $\{1, 3\}$     $\{2, 3\}$                        $\{1\}$     $\{2\}$     $\{3\}$     $\emptyset$

For  $|\Omega| = n$  the event space has  $2^n$  elements — which is exactly why the next unit is about counting.

# Not Every Sample Space Is Finite

## Dosage of a drug

$$\Omega = (0, \infty)$$

Uncountably many outcomes.

## Race with 7 teams

$$\Omega = \{\text{all orderings of } 1, \dots, 7\}$$

$7! = 5040$  outcomes.

Choosing  $\Omega$  is a modelling decision: it fixes what counts as an outcome, and therefore which questions can be asked at all.



# Venn Diagrams

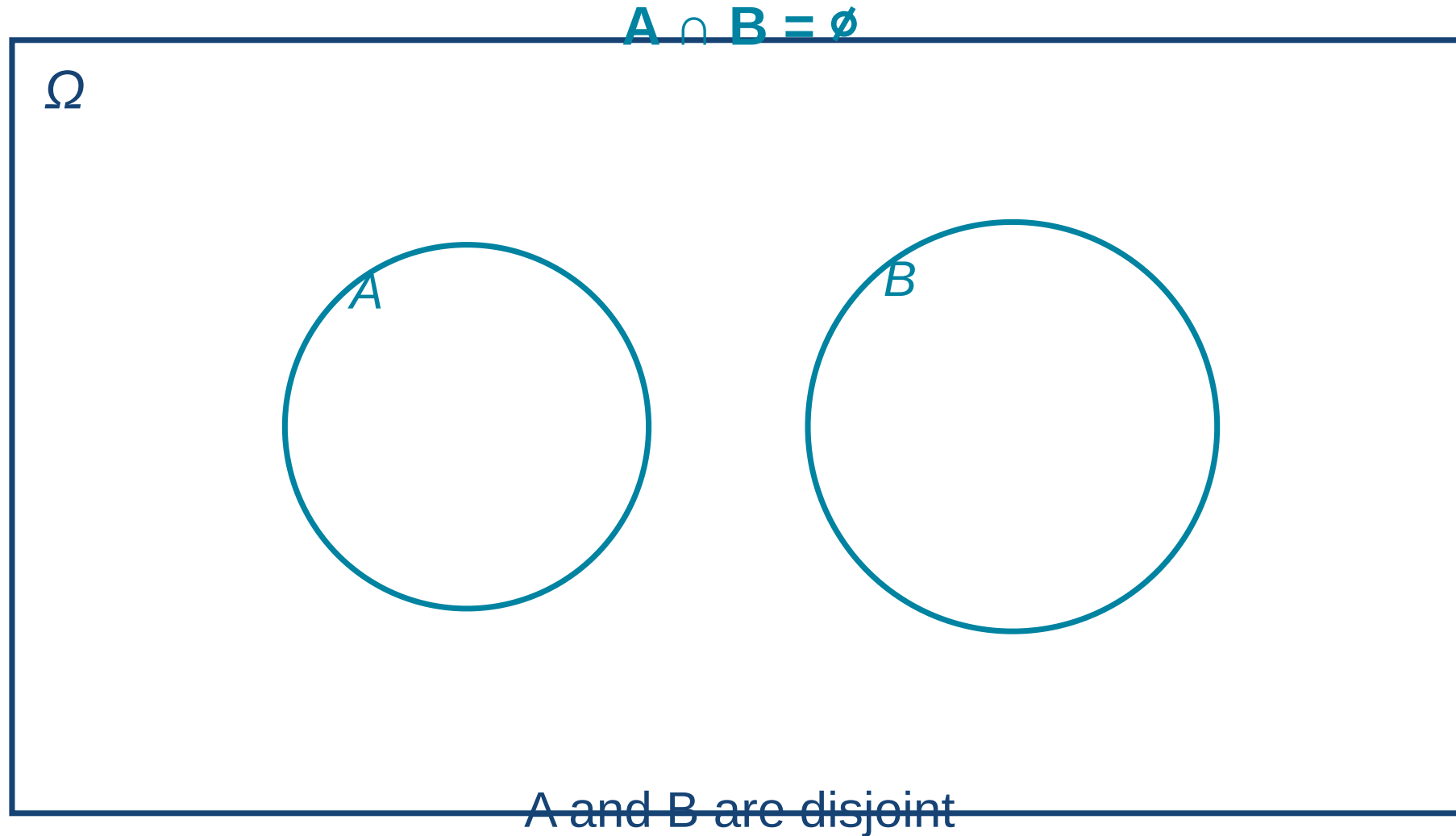
**Venn diagrams** are the graphical representation of events:

- the rectangle represents the sample space  $\Omega$
- circles represent events  $A \subseteq \Omega, A \in \mathcal{P}(\Omega)$

Every region of the picture *is* a set — and every set operation is a region you can point at.

# Important Event Types

Before each step: which region will light up?

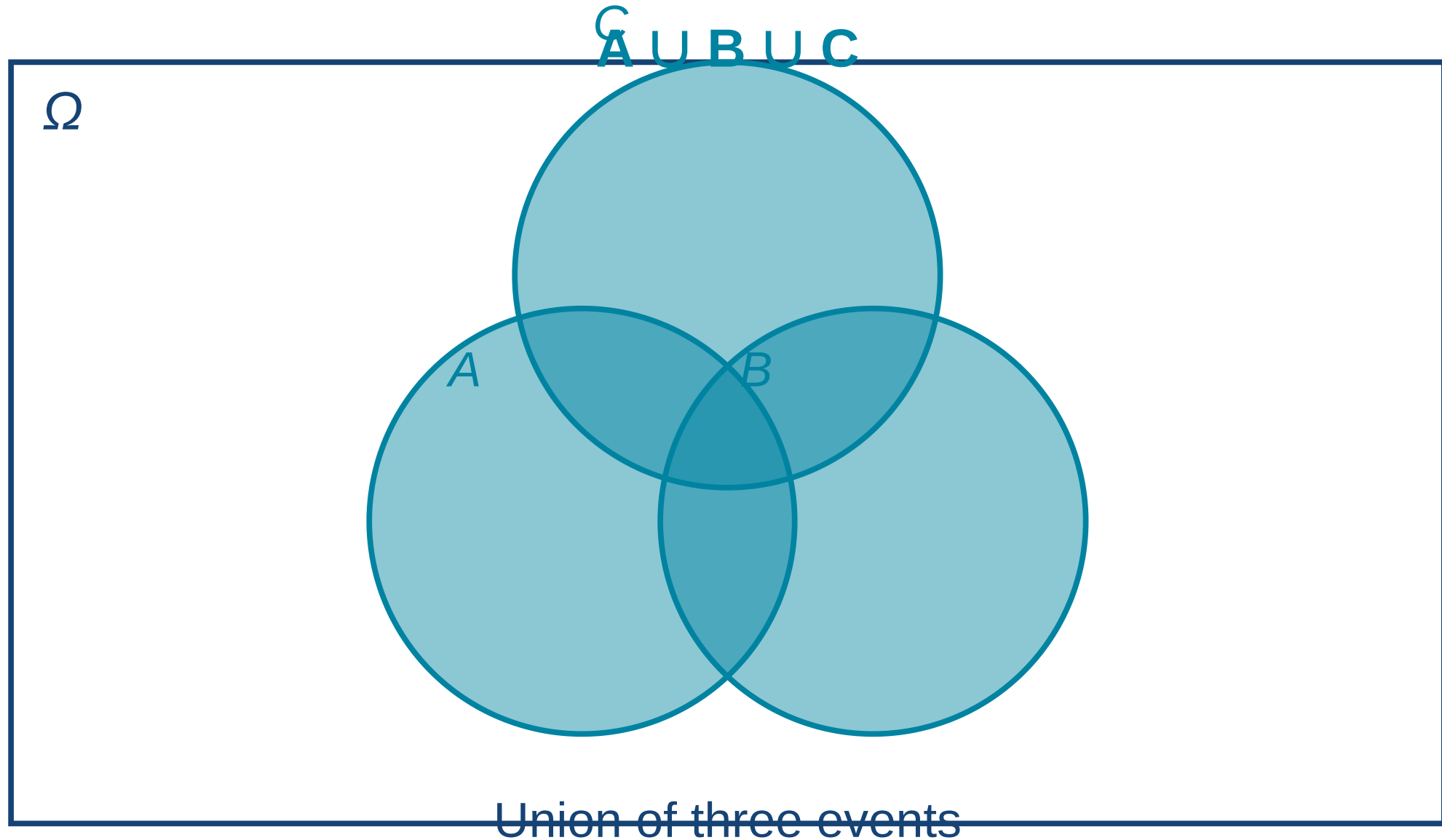


# Algebra of Events

| Law          | Union form                                       | Intersection form                                |
|--------------|--------------------------------------------------|--------------------------------------------------|
| Commutative  | $A \cup B = B \cup A$                            | $A \cap B = B \cap A$                            |
| Associative  | $(A \cup B) \cup C = A \cup (B \cup C)$          | $(A \cap B) \cap C = A \cap (B \cap C)$          |
| Distributive | $(A \cup B) \cap C = (A \cap C) \cup (B \cap C)$ | $(A \cap B) \cup C = (A \cup C) \cap (B \cup C)$ |

Events form a set algebra — the same laws you know, but now every identity is a picture.

# Set Operations as Regions



# Mini-Exercise

Sketch (pen and paper) the regions for:

1.  $(A \cup B) \cap C$
2.  $(A \cap B) \cup C$

Are they the same region? Which algebra law says so — or says they are not?

# Mini-Exercise — Solution

# Mini-Exercise — Solution

They differ:  $(A \cup B) \cap C$  lies entirely **inside**  $C$ , while  $(A \cap B) \cup C$  contains **all** of  $C$  plus the lens  $A \cap B$  outside it.

The distributive laws expand each into a different expression — the algebra distinguishes exactly what the picture shows.

# An Intuition: Probability as Area

Draw a Venn diagram on the ground, with each area proportional to the number of elementary events it contains. Throw a stone into the rectangle at random.

The probability of  $A$  is how likely the stone lands inside  $A$ , assuming

1. we hit the rectangle  $\Omega$  for certain, and
2. every part of  $\Omega$  is equally likely to be hit.

Probability is then a **measure of the state**  $A \subseteq \Omega$  of a random experiment — literally, how much of the sample space the event occupies.

Both assumptions are doing real work.  
Assumption 2 is the one we will have to give up.



# Two Ways to Read a Probability

## Frequentist

The probability of an elementary event  $a$  is the fraction of repetitions of the experiment that yield  $a$ .

*Requires the experiment to be repeatable.*

## Bayesian

The probability of  $a$  is the degree of certainty that  $a$  will happen — even without conducting the experiment.

*Requires a stated state of knowledge.*

Both interpretations yield the same mathematics. They are mental models for what the number *means*, not different theories.

# The Classical (Laplace) Definition

**Definition (Classical / Laplace probability).** If every elementary event in  $\Omega$  is assigned the same probability, then

$$P(A) = \frac{|A|}{|\Omega|} = \frac{\text{number of cases favourable for } A}{\text{number of possible outcomes}}$$

An experiment whose configuration and procedure make outcomes equiprobable — by definition or by construction — is a **Laplace experiment** its probabilities need not be estimated, they are implied by the setup.

# Why That Definition Is Not Enough

**The circularity.** "Laplace probability" defines probability using *equiprobability* — that is, using another notion of probability. Strictly speaking it is a circular definition.

The frequentist repair — defining probability as the limit of relative frequencies — fails too: such a limit exists only in mathematical settings, never in an experiment that can actually be repeated.

**So what do we do?** Stop trying to *define* what probability is, and instead specify what it must *do*.

# The Kolmogorov Axioms

**Definition (Probability measure, Kolmogorov 1933).** Let  $\Omega$  be a sample space and  $\mathcal{P}(\Omega)$  its event space. A function  $P : \mathcal{P}(\Omega) \rightarrow \mathbb{R}$  is a **probability measure** if:

## Axiom I

$$P(A) \geq 0$$

non-negative

## Axiom II

$$P(\Omega) = 1$$

normalised

## Axiom III

$$A \cap B = \emptyset \Rightarrow$$

$$P(A \cup B) = P(A) + P(B)$$

additive

Nothing is said about *which* probabilities the events get — only about how any assignment must behave.

**Beyond finite sample spaces.** When  $\Omega$  is infinite, the event space is not the full power set but a  $\sigma$ -**algebra** of measurable sets, and Axiom III is required for **countably many** disjoint

# Probability Space

**Definition.** Let  $\Omega$  be a sample space,  $\mathcal{P}(\Omega)$  an event space, and  $P$  a probability measure. The triple  $(\Omega, \mathcal{P}(\Omega), P)$  — also written  $(\Omega, P)$  — is a **probability space**.

Every probabilistic statement in this course is made inside some probability space. Naming it is the discipline that keeps modelling assumptions visible.

# What the Axioms Already Give Us

$$P(A) + P(\overline{A}) = 1$$

*(Axioms II, III)*

$$P(\emptyset) = 0$$

*(from the above with  $A = \Omega$ )*

$$A \subseteq B \Rightarrow P(A) \leq P(B)$$

monotonicity *(Axioms I, II)*

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

inclusion–exclusion *(Axiom III)*

For mutually exclusive  $A_1, \dots, A_p$ :  $P(A_1 \cup \dots \cup A_p) = P(A_1) + \dots + P(A_p)$ .

# Takeaway

Probability is not defined by counting or by repetition — it is any assignment of numbers to events that is **non-negative, normalised, and additive**. The Laplace formula is one such assignment, not the definition.

# Module 2

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## Conditioning Is Inference

**Guiding Question:** *A 99%-accurate test says you are sick. Why is that not a 99% chance of being sick — and what exactly does the evidence do to your belief?*



# Motivating Example

You roll a die twice. **After the first roll you see a 3.** Knowing this, what is the probability that both rolls sum to 8?

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You roll a die twice. **After the first roll you see a 3.** Knowing this, what is the probability that both rolls sum to 8?

- Before the first roll:  $\Omega = \{(1, 1), (1, 2), \dots, (6, 6)\}$ ,  $|\Omega| = 36$
- After seeing the 3, only  $\Omega_{red} = \{(3, 1), \dots, (3, 6)\}$  remains
- Outcomes like  $(2, 4)$  have become **impossible**
- The reduced space now carries all the probability:  $P(\Omega_{red}) = 1$
- "Sum is 8" inside it:  $E = \{(3, 5)\}$

$$P = \frac{1}{6}$$

# Definition

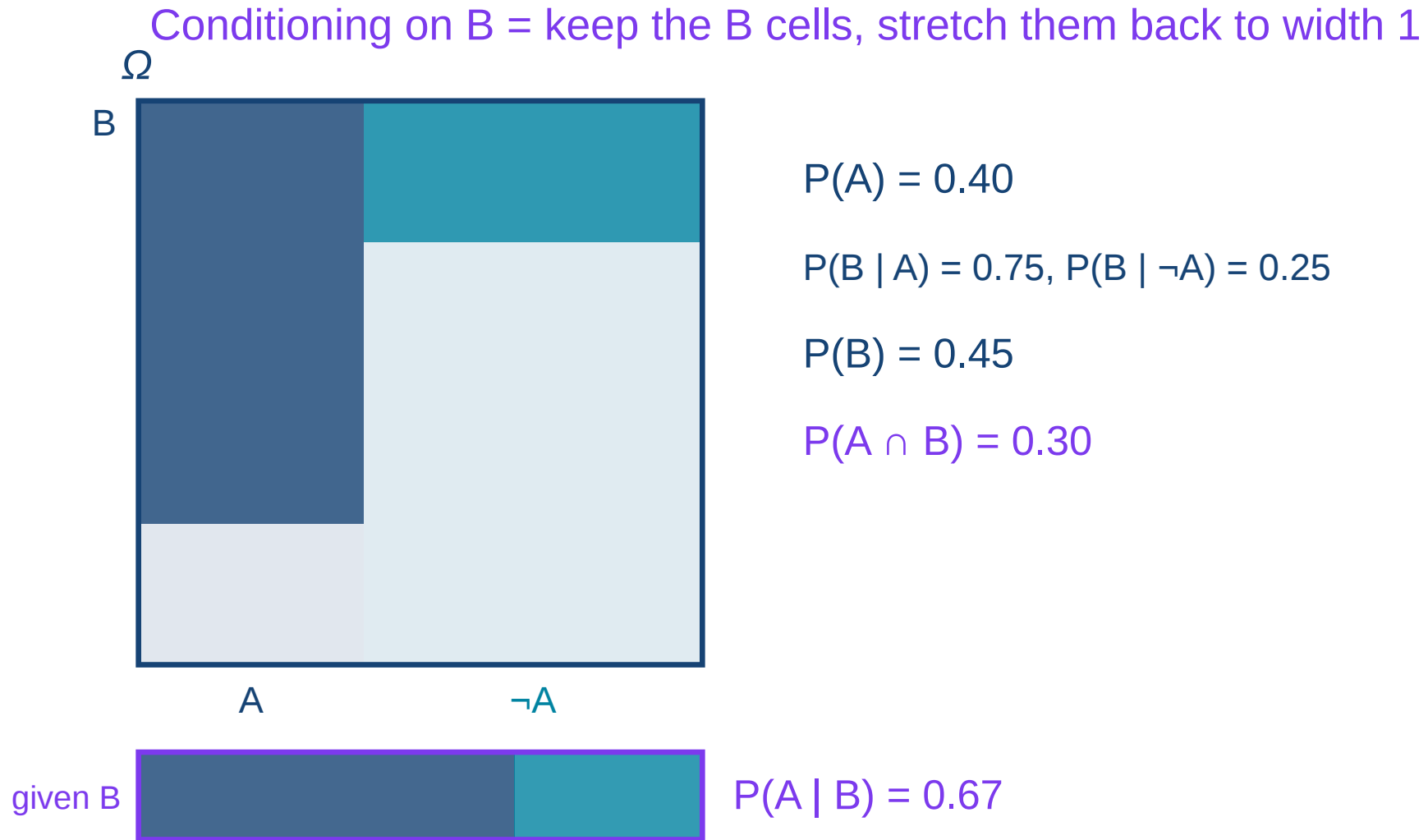
**Definition (Conditional probability).**

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}, \quad \text{if } P(B) > 0$$

$P(A \mid B)$  is the probability of  $A$  **under the condition**  $B$ .

Back to the stone: *given that the stone landed in area  $B$ , how likely is it that it also landed in  $A$ ?*  
We renormalise by the area of  $B$ , because  $B$  is now the whole world.

# Conditioning, Drawn



**Before the last click:** what must the two shaded cells be divided by?

# Example: Two Coins

Two coins are flipped. The event of interest is  $B = \{(h, h)\}$ .

**(a) "The first coin is heads"**

$$F = \{(h, h), (h, t)\}$$

$$P(B \mid F) = \frac{P(B \cap F)}{P(F)} = \frac{1/4}{1/2} = \frac{1}{2}$$

**(b) "At least one coin is heads"**

$$A = \{(h, h), (h, t), (t, h)\}$$

$$P(B \mid A) = \frac{P(B \cap A)}{P(A)} = \frac{1/4}{3/4} = \frac{1}{3}$$

Same event  $B$ , two different pieces of information, two different answers. **Why is (b) smaller?**

# Total Probability

**Theorem (Total probability).** Let  $B_1, \dots, B_k$  be mutually exclusive with  $\Omega = B_1 \cup \dots \cup B_k$  and  $P(B_i) > 0$ . Then for any event  $A$ :

$$P(A) = \sum_{i=1}^k P(B_i) \cdot P(A \mid B_i)$$

**Proof.** Decompose  $A$  along the partition and use additivity:

$$P(A) = P((B_1 \cup \dots \cup B_k) \cap A) = \sum_{i=1}^k P(B_i \cap A) = \sum_{i=1}^k P(B_i) \cdot P(A \mid B_i)$$

# Rules That Follow

## Product rule

$$\begin{aligned}P(A \cap B) &= P(B) \cdot P(A \mid B) \\ &= P(A) \cdot P(B \mid A)\end{aligned}$$

## Reversing the condition

$$P(B \mid A) = \frac{P(B) \cdot P(A \mid B)}{P(A)}$$

## Complement, same condition

$$P(\overline{A} \mid B) = 1 - P(A \mid B)$$

## But in general

$$P(A \mid \overline{B}) \neq 1 - P(A \mid B)$$

Held fixed at  $B$ , the map  $A \mapsto P(A \mid B)$  satisfies the Kolmogorov axioms — conditioning yields a **new probability measure**  $P_B$ .

# Chain Rule

$$P(E_1 \cap E_2 \cap \dots \cap E_n) = P(E_1) \cdot P(E_2 \mid E_1) \cdot P(E_3 \mid E_1 \cap E_2) \cdots P(E_n \mid E_1 \cap \dots \cap E_{n-1})$$

By successive application of the product rule.

Any joint probability can be written as a chain of conditionals. Choosing a *good* order — and dropping conditions that do not matter — is exactly what graphical models will do later.



# Takeaway

Conditioning **shrinks the sample space and renormalises**. Everything else — product rule, total probability, Bayes, the chain rule — is bookkeeping on that single move.

# Independence

**Definition (Statistical independence).** Two events  $A$  and  $B$  are **independent** iff

$$P(A \cap B) = P(A) \cdot P(B)$$

If additionally  $0 < P(B) < 1$ , this is equivalent to

$$P(A \mid B) = P(A) \quad \text{and} \quad P(A \mid \overline{B}) = P(A)$$

Independence means: **learning  $B$  changes nothing about  $A$ .**

# The Classic Confusion

Statistical independence does not mean the events are disjoint.

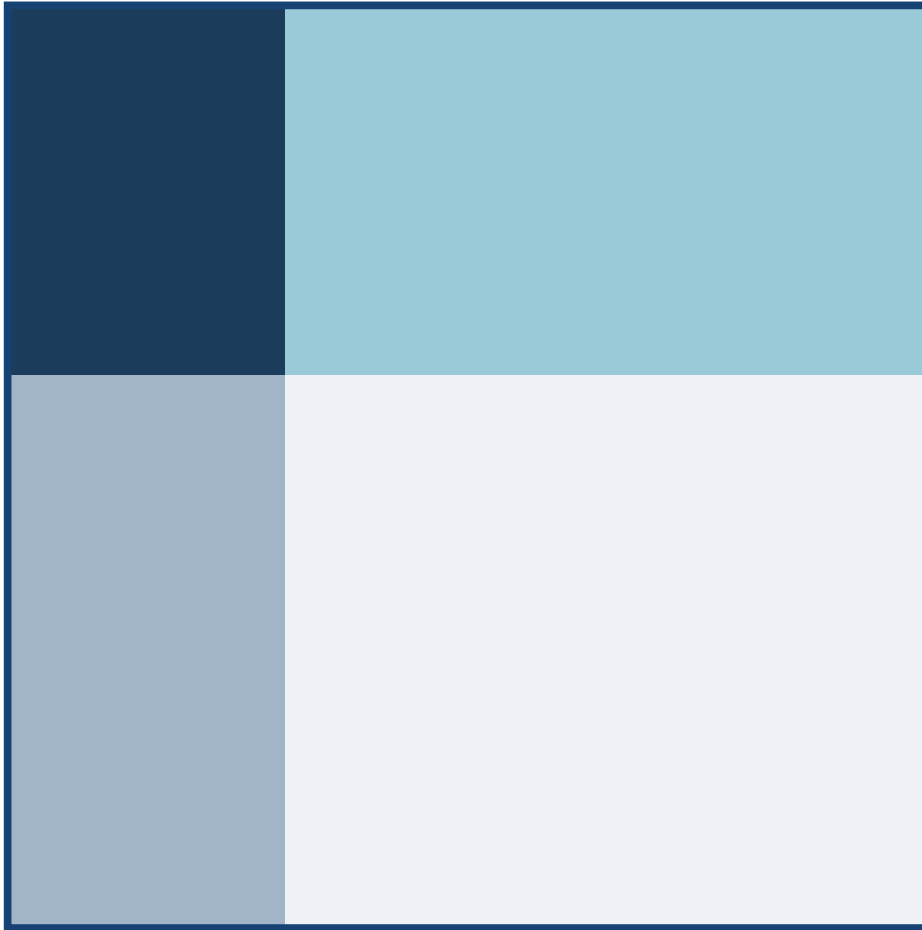
Disjoint events with positive probability are about as **dependent** as events can be: if  $A$  occurs,  $B$  certainly did not.

$$A \cap B = \emptyset \Rightarrow P(A \mid B) = 0 \neq P(A)$$

# Independence Is About Areas, Not Overlap

$$0.3 \cdot 0.4 = 0.12 = P(A \cap B) \rightarrow \text{independent}$$

$\Omega$



$$P(A) = 0.3$$

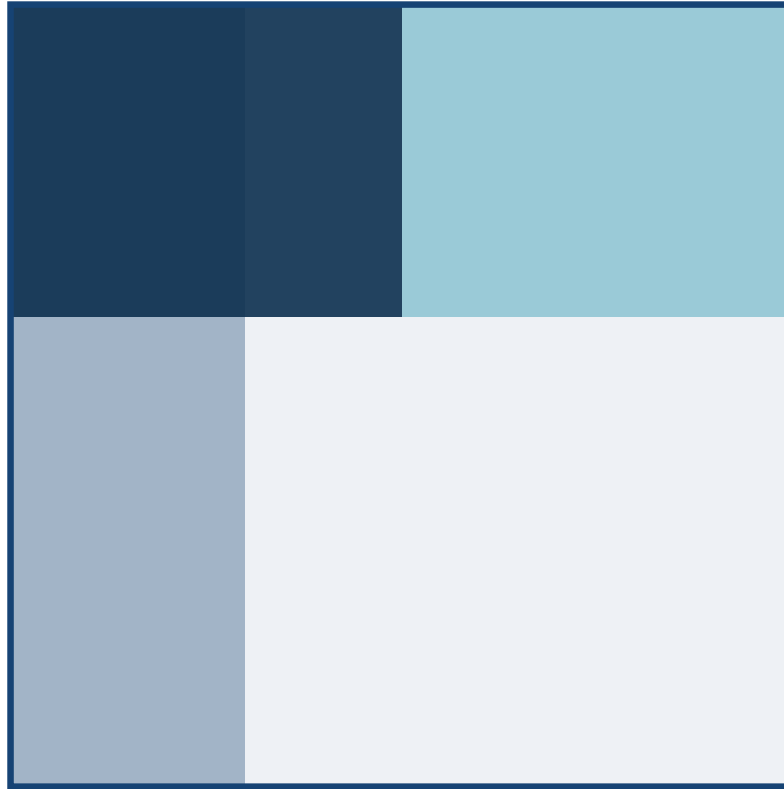
$$P(B) = 0.4$$

$$P(A \cap B) = 0.12$$

# The Same Picture, Dependent

$$0.3 \cdot 0.4 = 0.12 \neq 0.2 = P(A \cap C) \rightarrow \text{dependent}$$

$\Omega$



$$P(A) = 0.3$$

$$P(C) = 0.4$$

$$P(A \cap C) = 0.2$$

Here  $P(A) \cdot P(C) = 0.12$  but  $P(A \cap C) = 0.20$ : the intersection is **larger** than independence would predict —  $A$  and  $C$  occur together more often than chance.

# More Than Two Events

Events  $A_1, \dots, A_n$  are (mutually) independent iff for **every** subset of indices  $i_1, \dots, i_l$ :

$$P(A_{i_1} \cap \dots \cap A_{i_l}) = P(A_{i_1}) \cdots P(A_{i_l})$$

Pairwise independence is **not** enough — every subset must factorise.

# Takeaway

Independence is a statement about **numbers**, not pictures:  $P(A \cap B)$  equals the product of the marginals. Disjoint is not independent, and identical marginals can hide either.

# Bayes' Theorem — An Intuition First

A test for a rare disease is **99% accurate**: positive for 99% of the sick, negative for 99% of the healthy. About **1 in 1,000 people** has the disease.

**Your test comes back positive. How likely is it that you are actually sick?**



# Bayes' Theorem — An Intuition First

A test for a rare disease is **99% accurate**: positive for 99% of the sick, negative for 99% of the healthy. About **1 in 1,000 people** has the disease.

**Your test comes back positive. How likely is it that you are actually sick?**

Only about **9%** — the answer most people miss by a factor of ten.

# Why? Count People, Not Percentages

Restricting to the positives is exactly the conditioning step

$$P(\text{sick} \mid \text{positive}) = 1/11 \approx 9\%$$



# What Just Happened

Restricting attention to the people who tested positive is exactly the move from the conditional-probability definition:

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}$$

The 11 positives are the new sample space  $B$  the single sick person among them is  $A \cap B$ . The rarity of the disease survives the evidence because there were so many more healthy people to misclassify.

## The trap

"99% accurate" describes  $P(\text{positive} \mid \text{sick})$ .

You wanted  $P(\text{sick} \mid \text{positive})$ .

These are different numbers.

# Bayes' Theorem — Two Events

**Theorem (Bayes, 2 events).** Let  $(\Omega, \mathcal{P}(\Omega), P)$  be a probability space and  $A, B \in \mathcal{P}(\Omega)$  two events with  $P(B) > 0$ . Then:

$$P(A | B) = \frac{P(B | A) \cdot P(A)}{P(B)}$$

posterior — what we believe after      prior — belief before the evidence  
likelihood      evidence      how well A explains B      how well B explains A

Which of the four was "99% accurate", and which one did the intuition ignore?

# Proof — Two Events

From the definition of conditional probability, twice:

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)} = \frac{\frac{P(A \cap B)}{P(A)} \cdot P(A)}{P(B)} = \frac{P(B \mid A) \cdot P(A)}{P(B)}$$

# Bayes' Theorem — Multiple Hypotheses

**Theorem (Bayes, multiple events).** Let  $A_i \in \mathcal{P}(\Omega)$  be mutually exclusive with  $\Omega = A_1 \cup \dots \cup A_k$  and  $P(A_i) > 0$ , and let  $B$  be an event with  $P(B) > 0$ . Then

$$P(A_i | B) = \frac{P(B | A_i) \cdot P(A_i)}{P(B)} = \frac{P(B | A_i) \cdot P(A_i)}{\sum_{j=1}^k P(B | A_j) \cdot P(A_j)}$$

The denominator is the **total probability** of the evidence: the same observation, explained by every competing hypothesis, weighted by how much we believed each one beforehand.

# Proof — Multiple Hypotheses

As before, with the theorem of total probability supplying the denominator:

$$P(B) = \sum_{j=1}^k P(B \mid A_j) \cdot P(A_j)$$

# Mini-Exercise

Same test, but now screening a **high-risk group** where 1 in 20 people is sick — the test is unchanged.

Estimate  $P(\text{sick} \mid \text{positive})$  before computing it.



# Mini-Exercise — Solution

# Mini-Exercise — Solution

Out of 1,000: 50 sick  $\rightarrow \approx 50$  true positives 950 healthy  $\rightarrow \approx 10$  false alarms.

$$P(\text{sick} \mid \text{positive}) \approx \frac{50}{60} \approx 83$$

**The test did not change. The prior did.** This is why screening decisions depend on who is screened.

# Takeaway

Bayes' theorem is the arithmetic of changing your mind: **posterior**  $\propto$  **likelihood**  $\times$  **prior**.  
Evidence never speaks alone — it is always weighed against how plausible the hypothesis was to begin with.

# Module 3

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## From Outcomes to the Joint

**Guiding Question:** *We rarely care which outcome occurred, only what it was worth — and never about one quantity alone. How do we get from a sample space to the joint distribution the three moves start from?*

# Motivation: We Rarely Care About the Outcome Itself

Toss three fair coins:  $\Omega = \{H, T\} \times \{H, T\} \times \{H, T\}$ , eight outcomes.

Usually the interesting quantity is not *which* sequence occurred, but the **number of heads**  $Y$ :

|                          |                                   |                                   |                          |
|--------------------------|-----------------------------------|-----------------------------------|--------------------------|
| $P(Y = 0) = \frac{1}{8}$ | $P(Y = 1) = \frac{3}{8}$          | $P(Y = 2) = \frac{3}{8}$          | $P(Y = 3) = \frac{1}{8}$ |
| $(T, T, T)$              | $(H, T, T), (T, H, T), (T, T, H)$ | $(H, H, T), (T, H, H), (H, T, H)$ | $(H, H, H)$              |

$Y$  compresses eight outcomes into four numbers — and the probabilities come along automatically.

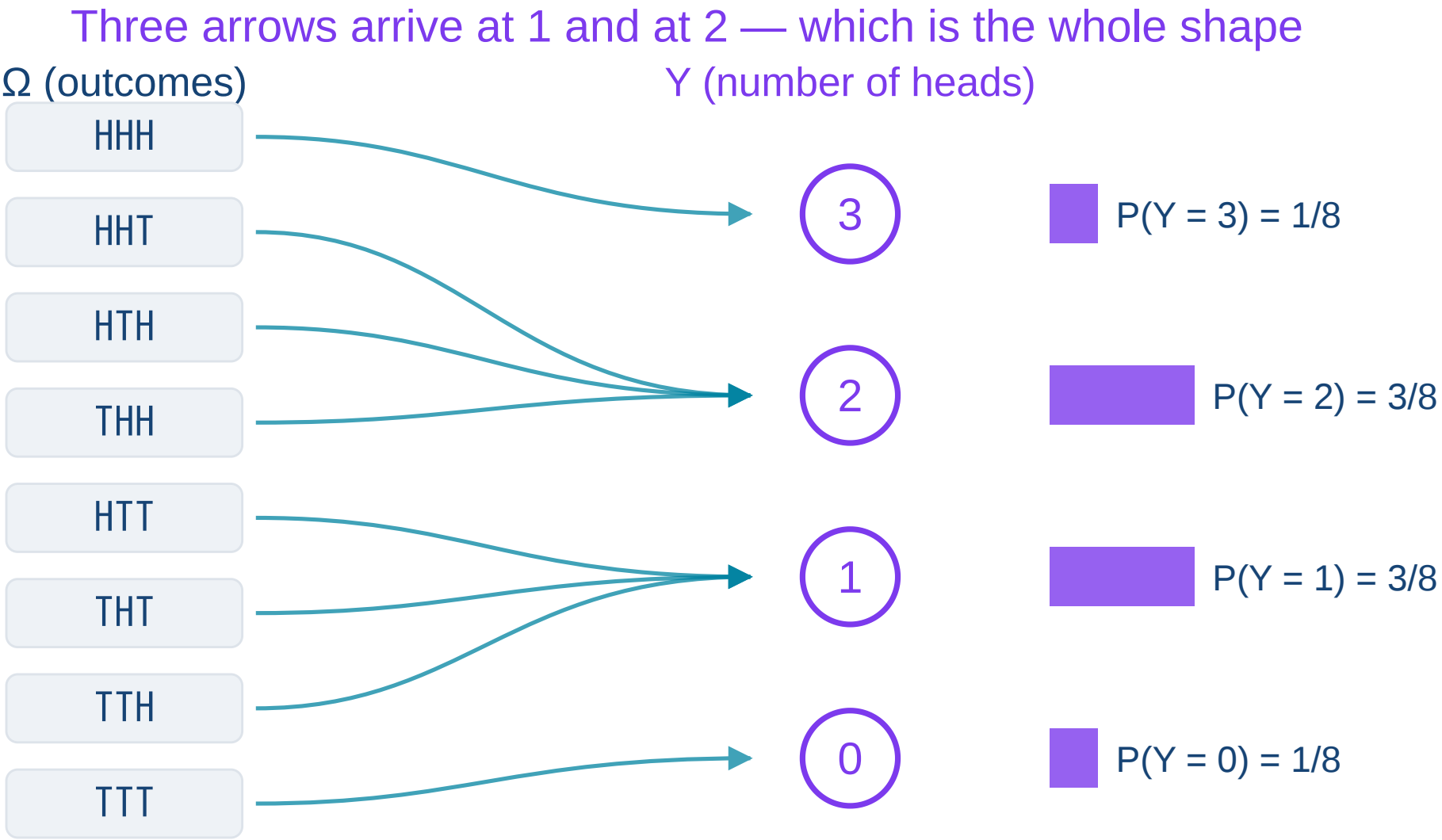
# Definition

**Definition (Random variable).** A random variable  $X$  maps outcomes to a measurable range  $E$ :  $X : \Omega \rightarrow E$ . Its probabilities are inherited from the underlying probability space:

$$P(X \in S) = P(\{\omega \in \Omega \mid X(\omega) \in S\})$$

A random variable is neither random nor a variable — it is a **function** on the sample space. The randomness lives in  $\omega$ , not in  $X$ .

# The Function, Drawn



# Examples

## Two coins

$$\Omega = \{HH, HT, TH, TT\}$$

$X$  = number of heads

## People

$$\Omega = \{\text{person}_1, \text{person}_2, \dots\}$$

$X$  = height of the person

## Two dice

$$\Omega = \{\langle n_1, n_2 \rangle\}$$

$$X(\omega) = n_1 + n_2$$



# Discrete and Continuous

## Discrete random variables

Countable range, usually  $E \subseteq \mathbb{N}$ .

- number of children of a person
- sum of two dice

## Continuous random variables

Real-valued range,  $E = \mathbb{R}$ .

- height of a person
- horsepower of a car

The distinction decides *how* we may describe the probabilities — the subject of the next two units.

# A Random Variable With Infinitely Many Values

Flip a biased coin ( $P(H) = p$ ) until the first head. Let  $X$  be the number of flips needed:

$$P(X = 1) = p, \quad P(X = 2) = (1 - p)p, \quad P(X = 3) = (1 - p)^2 p$$

$$P(X = n) = (1 - p)^{n-1} p$$

Countably infinite range, and the probabilities still sum to 1.

# Takeaway

A random variable is a **function from outcomes to numbers** that carries the probability measure with it. It lets us stop talking about sample spaces and start talking about quantities — which is what the rest of the course does.

# Probability Mass Function

**Definition (PMF).** For a discrete random variable  $X$ , the **probability mass function** is

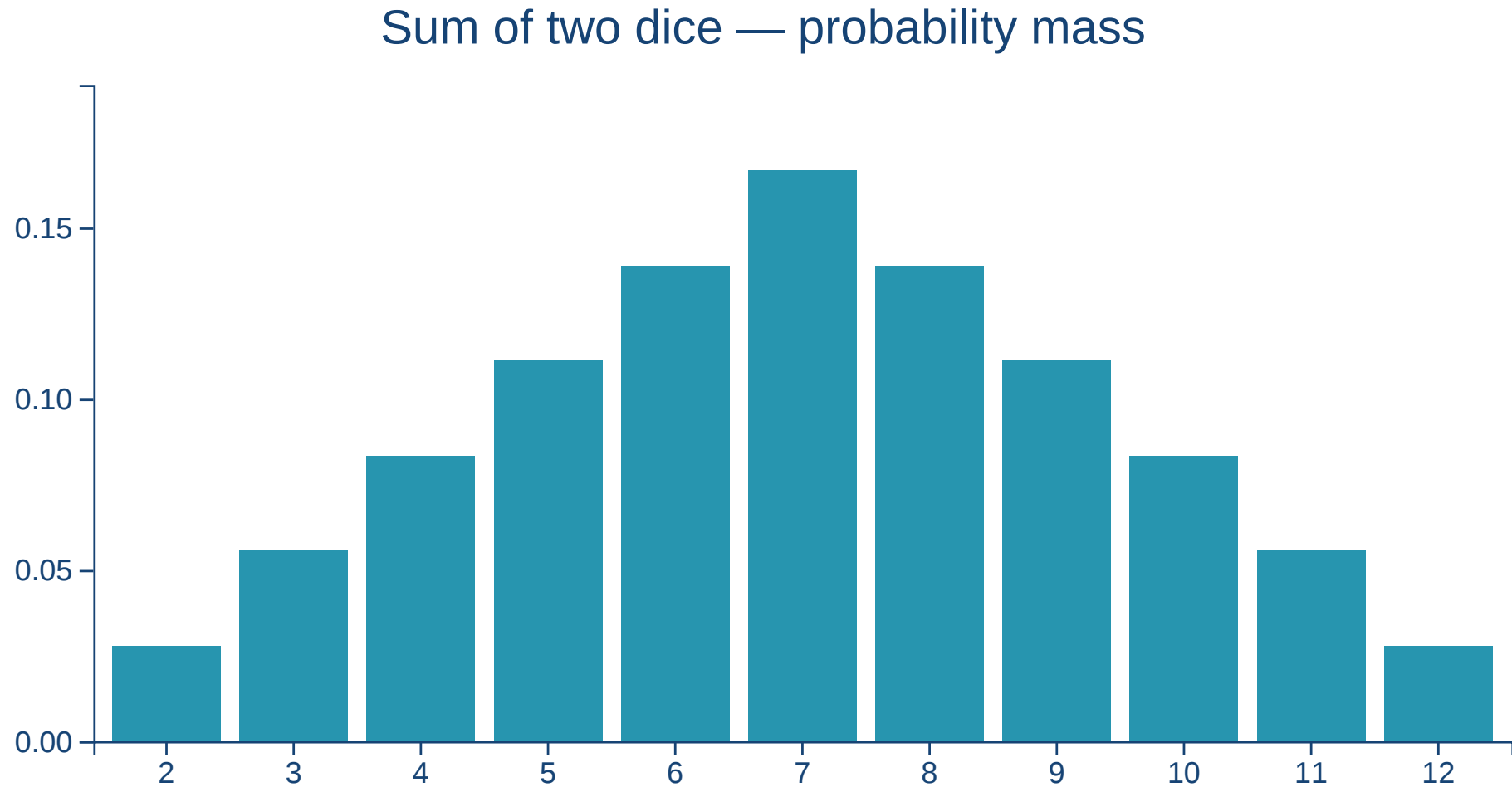
$$p(a) = P(X = a)$$

$p(a) > 0$  for at most countably  
many  $a$

$$\sum_i p(a_i) = 1$$

$p$  is itself a probability measure

# Example: Sum of Two Dice



The triangular shape is the counting from earlier units, drawn: six of the 36 outcomes give 7, only one gives 2.

# Expectation

Definition (Expected value).

$$E[X] = \sum_i a_i \cdot p(a_i)$$

- Frequentist reading: the **average outcome** if the experiment were repeated infinitely often
- The first moment:  $\mu = E[X^1]$  the first *central* moment  $E[X - E[X]]$  is always 0

For a real function  $g$ :

$$E[g(X)] = \sum_i g(a_i)p(a_i)$$

and for constants  $a, b$ :

$$E[aX + b] = aE[X] + b$$

# Variance

## Definition (Variance).

$$\text{Var}(X) = E[(X - \mu)^2] = E[X^2] - (E[X])^2$$

- a measure of **dispersion** around the mean — larger variance, larger typical deviation
- the second central moment
- $\text{Var}(aX + b) = a^2 \text{Var}(X)$
- $\sigma = \sqrt{\text{Var}(X)}$  is the **standard deviation**, in the same unit as  $X$

# Bernoulli and Binomial

**Bernoulli**( $p$ ) — one trial

$X = 1$  success,  $X = 0$  failure

$$E[X] = p, \quad \text{Var}(X) = p(1 - p)$$

**Binomial**  $B(n, p)$  —  $n$  independent trials

$$p(k) = \binom{n}{k} p^k (1 - p)^{n-k}$$

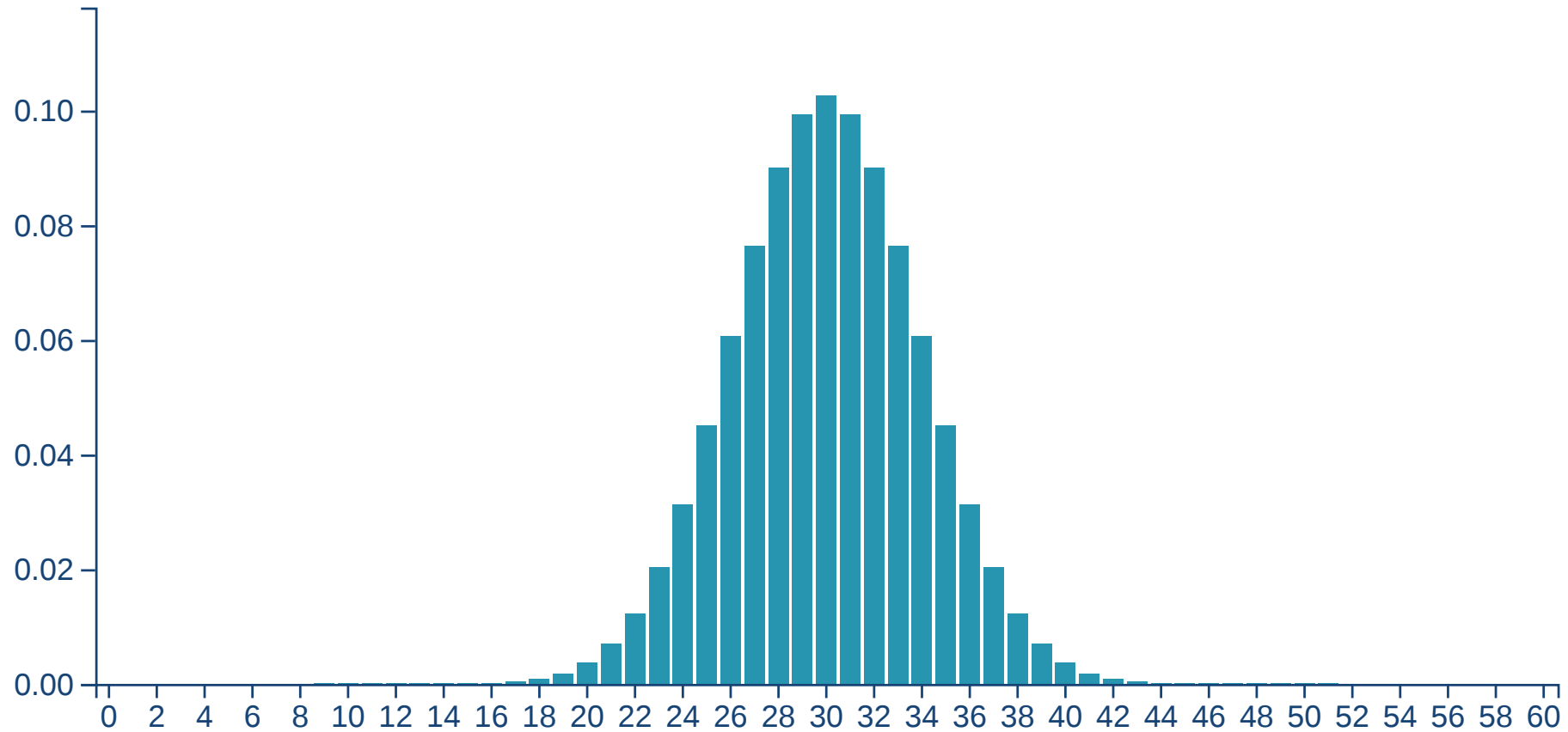
$$E[X] = np, \quad \text{Var}(X) = np(1 - p)$$

The binomial PMF is the counting formula times the probability of one specific sequence — combinatorics and independence in a single line.



# How the Shape Depends on the Parameters

Binomial  $B(n = 60, p = 0.5)$  — probability mass



**Predict before each click:** as  $n$  grows, does the distribution get wider or narrower — and what happens to its shape?

# Mini-Exercise

A quiz has 10 questions with 4 options each. A student guesses every answer.

1. Which distribution describes the number of correct answers?
2. What is the expected number correct?
3. Is passing ( $\geq 5$  correct) likely?

# Mini-Exercise — Solution

# Mini-Exercise — Solution

1.  $B(10, 0.25)$  — ten independent trials, constant success probability
2.  $E[X] = 10 \cdot 0.25 = 2.5$
3.  $P(X \geq 5) \approx 0.078$  — under 8%

The independence assumption is what makes this a binomial ask when it would fail in a real exam.

# Takeaway

A discrete distribution is a PMF expectation and variance compress it into **where it sits** and **how much it spreads**. Bernoulli and binomial are the counting units in probabilistic clothing.

# From Mass to Density

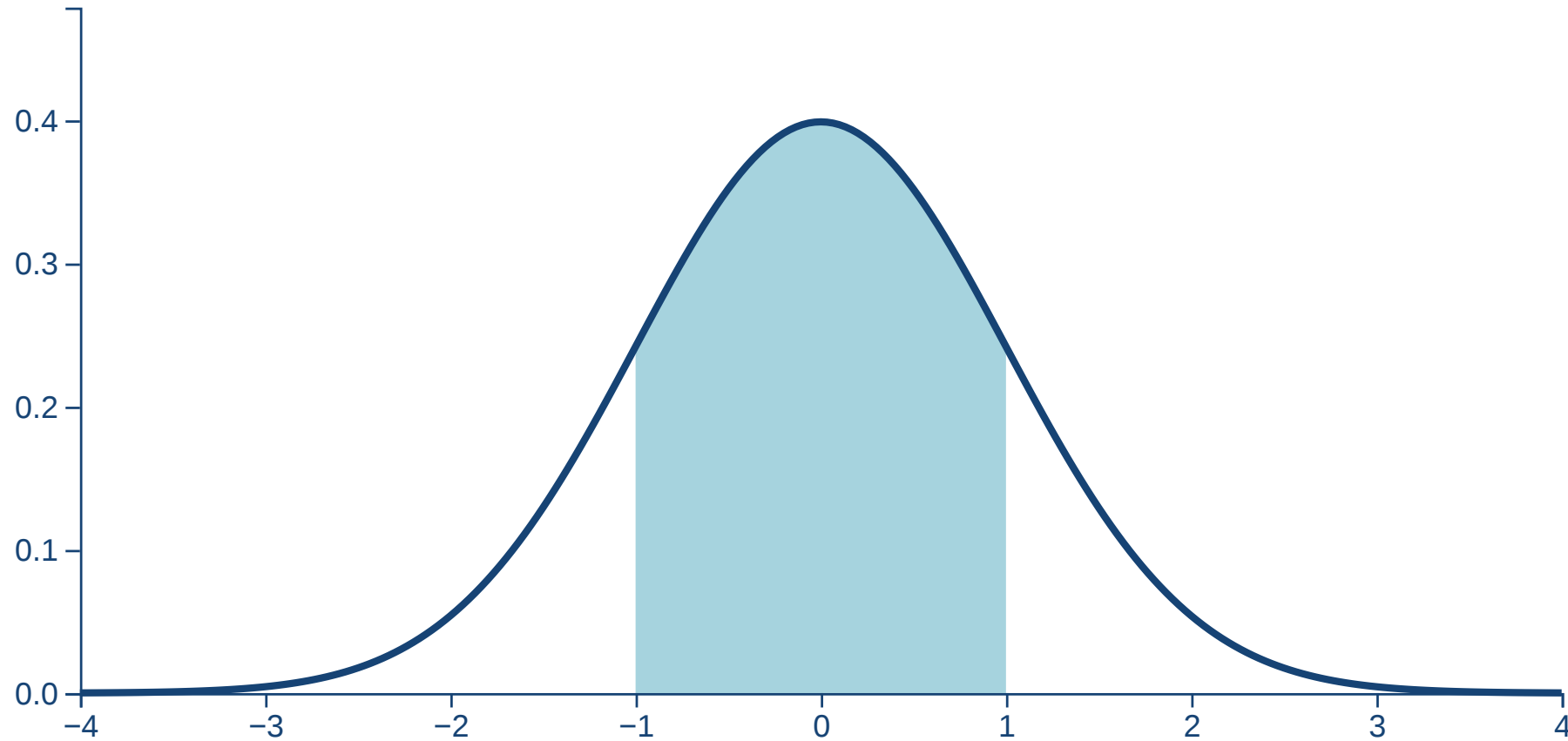
**Definition (PDF).** A continuous random variable  $X$  has a **probability density function**  $f$  with

$$P(b \leq X \leq c) = \int_b^c f(a)da, \quad \int_{-\infty}^{\infty} f(a)da = 1$$

$f(a)$  is **not** a probability — it is probability *per unit interval*. Consequently  $f(a) > 1$  is perfectly possible, and  $P(X = a) = 0$  for every single value  $a$ .

# Probability Is Area Under the Curve

Normal  $N(\mu = 0, \sigma = 1)$  — density



The shaded region is  $P(-1 \leq X \leq 1)$  — the same "probability as area" intuition from the axioms of probability, now with a curve instead of a Venn diagram.

# Expectation and Variance — Sums Become Integrals

## Discrete

$$E[X] = \sum_i a_i p(a_i)$$

$$E[g(X)] = \sum_i g(a_i) p(a_i)$$

## Continuous

$$E[X] = \int_{-\infty}^{\infty} a f(a) da$$

$$E[g(X)] = \int_{-\infty}^{\infty} g(a) f(a) da$$

The rules are unchanged:  $E[aX + b] = aE[X] + b$ , and  $\text{Var}(X) = E[(X - \mu)^2]$ .



# Two Basic Continuous Distributions

**Uniform on  $[a, b]$**

$$f(x) = \frac{1}{b-a} \text{ for } x \in [a, b]$$

Every interval of equal length is equally likely — the Laplace assumption, continuous version.

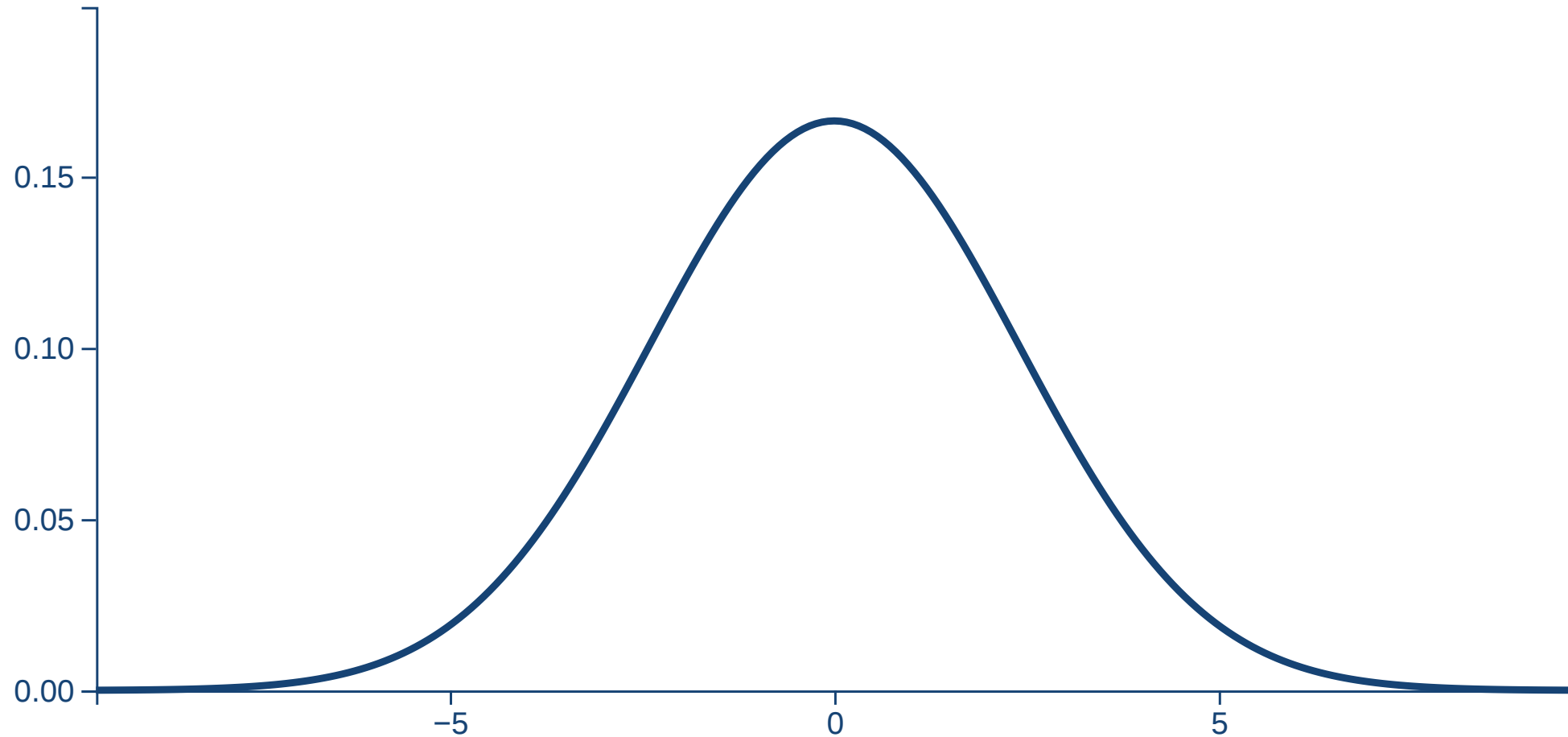
**Normal  $N(\mu, \sigma^2)$**

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

$E[X] = \mu$ ,  $\text{Var}(X) = \sigma^2$ ;  $N(0, 1)$  is the **standard** normal.

# What $\sigma$ Does to the Shape

Normal  $N(\mu = 0, \sigma = 2.4)$  — density



The peak falls as the curve widens — the area is always 1.

# Why the Normal Is Everywhere

The **central limit theorem**: under mild conditions, averages of many independent observations converge to a normal distribution — whatever the original distribution was.

This is why the normal is the default when the true distribution is unknown, and why the binomial started looking bell-shaped as  $n$  grew in the previous unit.

# Other Distributions You Will Meet

- **Exponential** — time until an event occurs (e.g. until the next earthquake)
- **Gamma** — time until  $n$  events have occurred
- **Beta** — values confined to a finite interval  $[a, b]$
- **Dirichlet** — distributions over probabilities

Continuous distributions replace mass with **density** and sums with **integrals**. Probability remains area, expectation remains a weighted average, and the normal is the default shape because of the central limit theorem.

# More Than One Random Variable

## Univariate

one random variable

## Bivariate

two random variables

## Multivariate

three or more

A **joint distribution** describes the outcomes of several random variables *at the same time* — which is what every dataset with more than one column is.

# Joint Mass and Density

**Discrete** — joint PMF

$$p_{X,Y}(a,b) = P(X = a, Y = b)$$

**Continuous** — joint PDF

$$P((X,Y) \in R) = \iint_R f_{X,Y}(a,b) da db$$

**Notational convention from here on.** We stop distinguishing discrete from continuous variables and simply write  $p(a,b,c,\dots)$  for the probability that  $\{X_1, X_2, X_3, \dots\}$  is in state  $a, b, c, \dots$

# Marginalisation

**Marginalisation** eliminates a random variable by summing over all its states. For  $p(x_1, \dots, x_n)$ , a **marginal** over a subset is

$$p(x_1) = \sum_{x_2} \cdots \sum_{x_n} p(x_1, x_2, \dots, x_n)$$

"Forget about a variable, but keep its influence correctly accounted for."

# Example: Boys and Girls

15% of families have no children, 20% have one, 35% have two, 30% have three. Each child is independently a boy or a girl with probability  $\frac{1}{2}$ .

Let  $B$  = number of boys,  $G$  = number of girls.

| $p(B, G)$ | $G = 0$ | $G = 1$ | $G = 2$ | $G = 3$ | $p(B)$ |
|-----------|---------|---------|---------|---------|--------|
| $B = 0$   | .15     | .10     | .0875   | .0375   | .375   |
| $B = 1$   | .10     | .175    | .1125   | 0       | .3875  |
| $B = 2$   | .0875   | .1125   | 0       | 0       | .2     |
| $B = 3$   | .0375   | 0       | 0       | 0       | .0375  |
| $p(G)$    | .375    | .3875   | .2      | .0375   | 1      |

The margins of the table *are* the marginals — which is where the name comes from.



# Covariance

$$\text{cov}(A, B) = E[(A - E[A])(B - E[B])]$$

$$\text{cov} > 0$$

$A$  large when  $B$  large —  
positively related

$$\text{cov} < 0$$

$A$  large when  $B$  small —  
negatively related

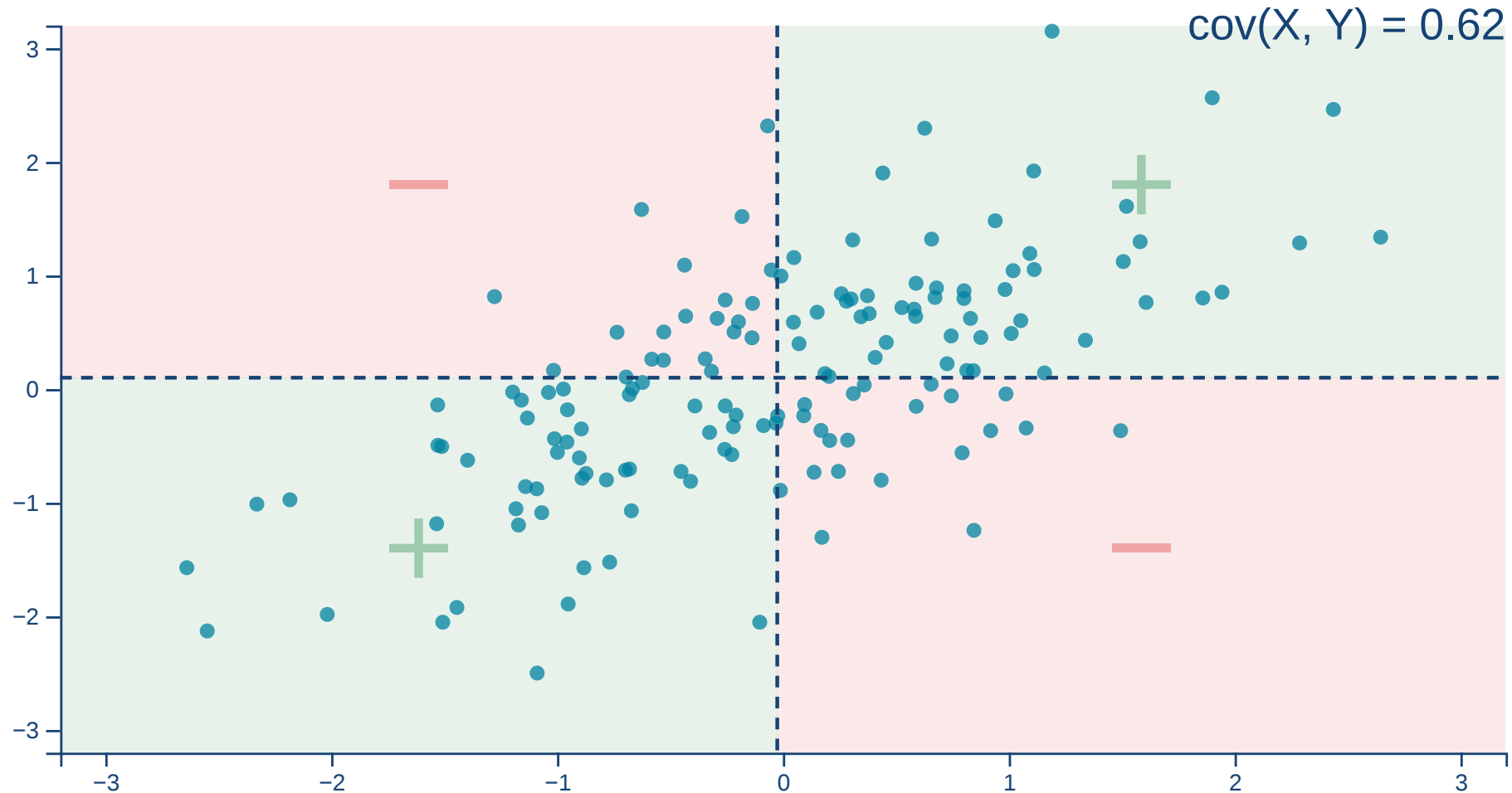
$$\text{cov}(A, A) = \text{Var}(A)$$

variance is self-covariance

Independence implies zero covariance. **The converse is false** — covariance only detects *monotone linear* association.

# Covariance Sees Only One Kind of Structure

Slide  $\rho$ , then switch the relation. Predict the covariance first.



**Predict first:** with  $Y = X^2 - 1$ , every point is fixed by its  $x$ . The covariance?

# Takeaway

The joint distribution holds everything **marginalisation** sums a variable away, **conditioning** restricts, and **covariance** summarises linear co-variation. Zero covariance is weaker than independence.

# Module 4

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## The Data Is a Sample

**Guiding Question:** *The distribution is unknown and all we ever have is a sample. What can we honestly say about the population — and how wrong are we allowed to assume we are?*

# We Never Know the Distribution

Up to now we assumed a distribution was given. In practice it is **unknown**, and we must estimate its properties from data.

## Population

The entire group — all humans on earth, all transactions, all documents.

Its properties  $(\mu, \sigma^2)$  are the truth we cannot observe.

## Sample

The observations we actually have.

Its properties  $(\bar{x}, s^2)$  are what we compute — estimates of the population's.

**Descriptive statistics** summarise a sample, usually while assuming something about the underlying distribution.

# What We Summarise

## Central tendency

1st order moment  
sample mean, median

## Dispersion

2nd central moment  
variance, standard deviation, IQR

## Shape

3rd central moment  
skewness, quantiles

Univariate statistics describe one random variable **multivariate** statistics describe  $X_1, \dots, X_d$  together.

# Means

**Sample mean**

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$$

**Weighted mean**

$$\bar{x}_w = \frac{\sum_i w_i x_i}{\sum_i w_i}$$

**Geometric mean** — for rates and ratios

$$\left( \prod_i x_i \right)^{1/n}$$

**Harmonic mean** — for averaging rates

$$\frac{n}{\sum_i 1/x_i}$$

**Trimmed mean:** drop the  $k$  smallest and largest values, then average — a cheap way to buy robustness.

# Median

The **median** is the middle value of the sorted sample: for odd  $n$  the  $\frac{n+1}{2}$ -th value, for even  $n$  the average of the two middle values.

**Robust** against outliers — one extreme value moves the median by at most one position

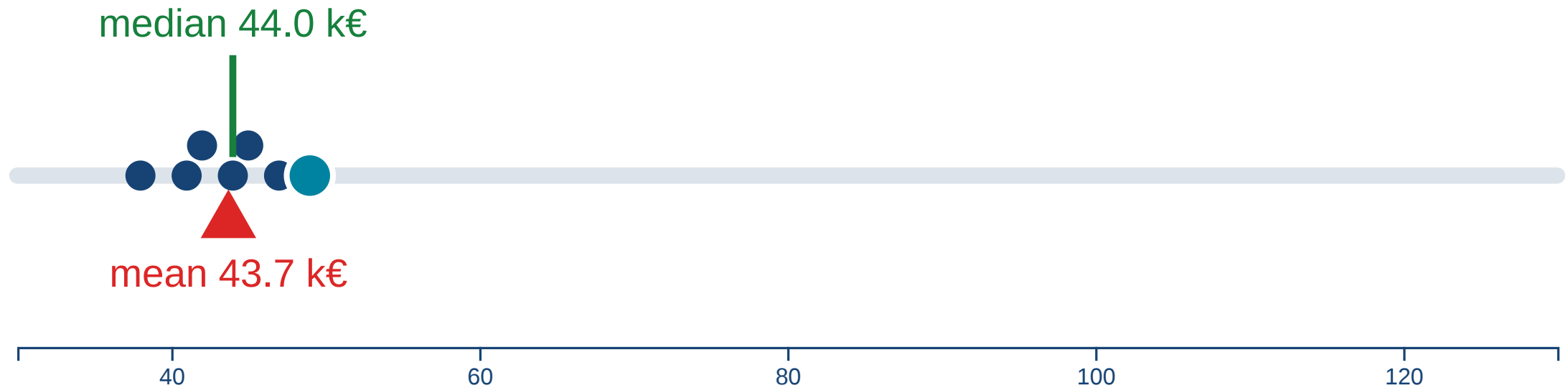
**Costlier** to compute: requires sorting, or an interval approximation over binned data

For binned data, the median is interpolated within its interval from the cumulative counts, the interval's lower border and width.



# One Value, Two Very Different Reactions

Drag the light point — which marker follows it?



**Commit first:** drag the top salary to the far right. Which marker follows?



# Mini-Exercise

Salaries in a small firm (in k€): 38, 41, 42, 44, 45, 47, 260.

Compute the mean and the median. Which number should appear in a recruitment ad, and which one describes a typical employee?

# Mini-Exercise — Solution

# Mini-Exercise — Solution

Mean  $\approx 73.9$ , median = 44.

The single large value pulls the mean above **every** salary except one. The median describes the typical employee the mean describes the payroll divided by headcount. Both are correct — they answer different questions.

# Dispersion

**Dispersion** describes how stretched or squeezed the data is around its centre.

**Sample variance** — the variance *of the observed data*

$$s^2 = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2$$

**Standard deviation**  $s = \sqrt{s^2}$ , in the unit of the data.

Computable in **one pass** over the data,  $O(n)$ , using

$$\sum x_i^2 \text{ and } \sum x_i$$

— which matters when the data does not fit in memory.

# Why the Denominator Changes

The **population variance**  $\sigma^2$  is the variance over the whole population. If the true mean  $\mu$  is unknown, we must estimate it — and using  $\bar{x}$  in its place makes the deviations systematically too small.

$$\hat{\sigma}^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2$$

Dividing by  $n - 1$  compensates: one degree of freedom was spent estimating the mean from the same data.

# Takeaway

$s^2$  describes **the sample**  $\hat{\sigma}^2$  estimates **the population**. The  $n - 1$  is not a convention — it is the price of having estimated the mean from the same data.

# Quantiles

The  $k$ -th  $q$ -**quantile** is the value  $x$  such that  $k/q$  of the values are smaller and  $(q - k)/q$  are larger ( $0 < k < q$ ).

## Percentiles

$$q = 100$$

"the 12th percentile"

## Quartiles

$$q = 4$$

$Q_1 = 25\text{th}$ ,  $Q_2 = 50\text{th}$  (= median),

$Q_3 = 75\text{th}$  percentile

## IQR

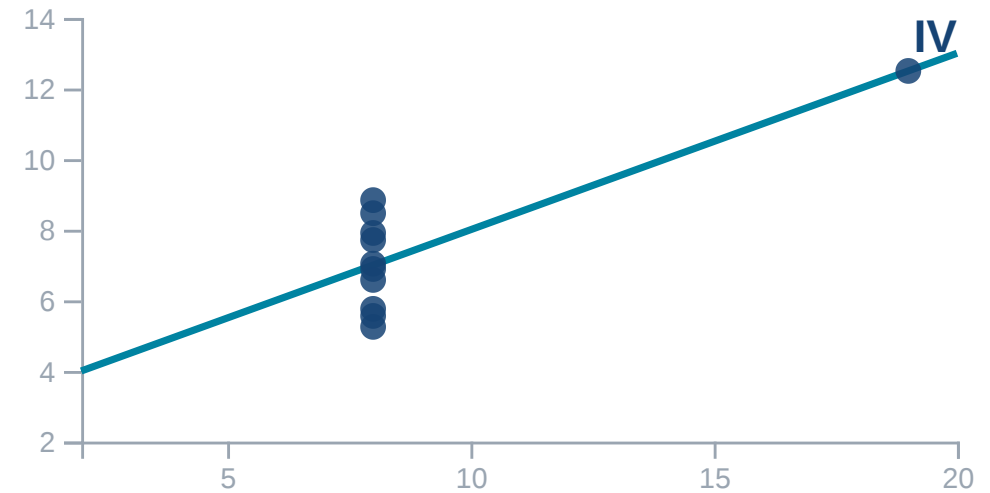
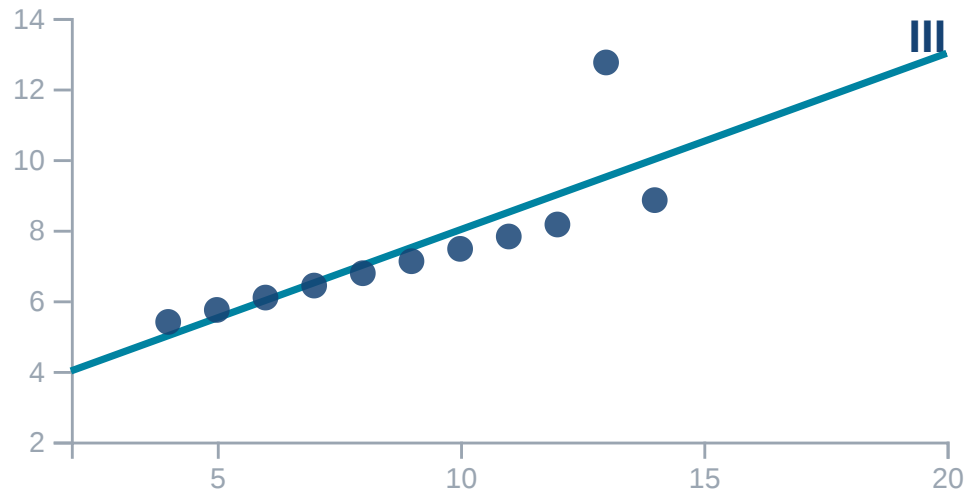
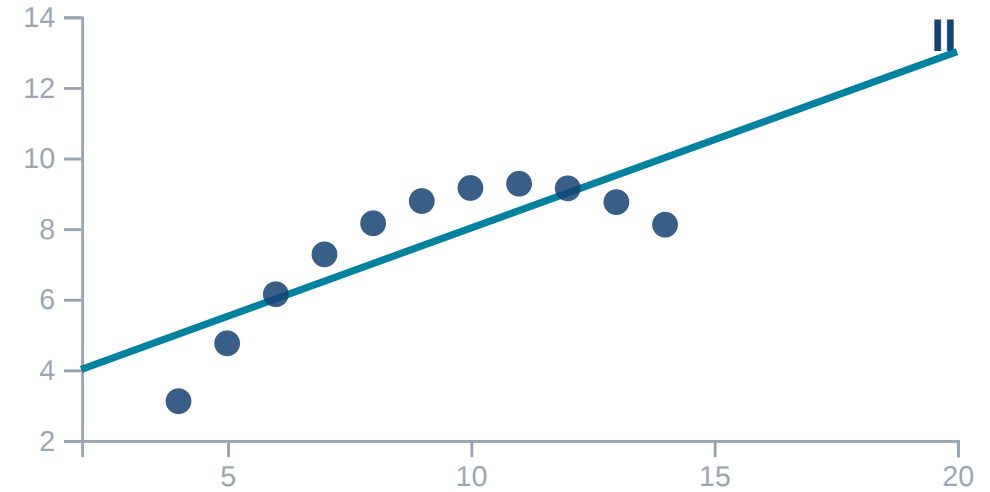
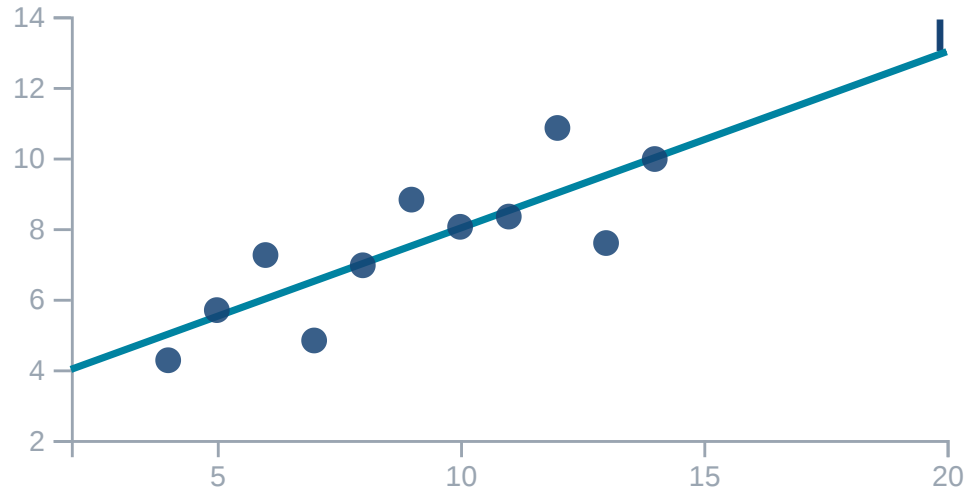
$$\text{IQR} = Q_3 - Q_1$$

the middle half of the data

Quantiles summarise **shape and skewness** without assuming a distribution.



# Anscombe's Quartet



Same mean, same variance, same correlation, same regression line — different data



# What the Quartet Teaches

**Summaries compress — and compression loses.**

Identical moments can hide a curve, an outlier, or a variable that barely varies at all.

**Plot first, summarise second.**

Visual inspection is not a soft alternative to statistics it is how you find out which statistics are meaningful.

Quantiles describe a distribution without assuming its shape — and no set of summary numbers can replace looking at the data.

# You Computed a Number. Which Number Did You Want?

Seven salaries gave a sample mean of  $\bar{x} = 43.7$  k€.

What is the probability that the population mean  $\mu$  is exactly 43.7 k€?

# You Computed a Number. Which Number Did You Want?

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What is the probability that the population mean  $\mu$  is exactly 43.7 k€?

Zero.  $\bar{x}$  is not  $\mu$  — it is one **estimate** of it, computed from the seven people who happened to be in the sample.

The honest question is not *what is  $\mu$*  but **how far from  $\mu$  can this number be?**

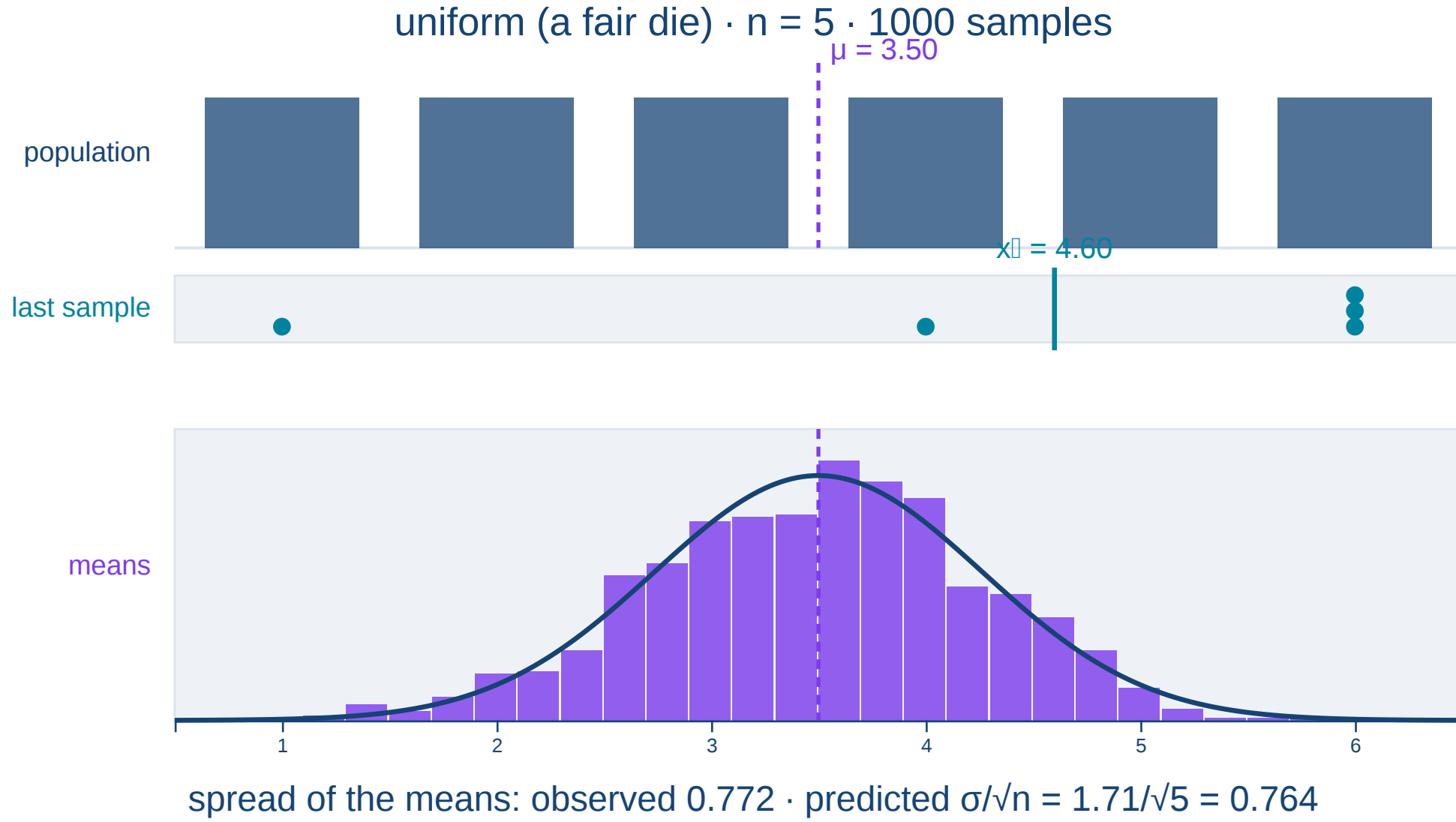
# Estimator and Estimate

**Definition.** An **estimator** is a function of the sample —  $\hat{\theta} = g(X_1, \dots, X_n)$  — used to approximate an unknown quantity  $\theta$  of the population. The number it takes on for one particular sample is an **estimate**.

The sample is random, so the estimator is a **function of random variables**:  $\hat{\theta}$  is itself a random variable, with a distribution of its own.

$\bar{x}$ ,  $s^2$  and  $\hat{\sigma}^2$  from the previous units are all estimators. We used them without saying what kind of object they are.

# An Estimator Has a Distribution



# Three Things the Picture Shows

- The means pile up **around**  $\mu$  — the estimator is not systematically off
- The pile is **narrower than the population** — averaging cancels noise
- The pile is **bell-shaped**, even though the population was not

**Definition (Sampling distribution).** The distribution of an estimator  $\hat{\theta}$  over all possible samples of size  $n$ .  $\hat{\theta}$  is **unbiased** if  $E[\hat{\theta}] = \theta$ .

# Use the Widget: Change the Population

Set the population to **skewed** and then to **bimodal**, keeping  $n = 5$ .

Which of the three observations survive a population that is not symmetric?



# Use the Widget: Change the Population

Set the population to **skewed** and then to **bimodal**, keeping  $n = 5$ .

Which of the three observations survive a population that is not symmetric?

All three. The shape of the sampling distribution stops depending on the shape of the population as soon as  $n$  is more than a handful — which is why one formula can cover cases we never inspect.

# Standard Error

**Definition (Standard error).** The standard deviation of an estimator's sampling distribution. For the sample mean of independent draws:

$$\text{SE}(\bar{X}) = \frac{\sigma}{\sqrt{n}}$$

It is measured in the unit of the data and answers "how far off is a typical estimate", where  $\sigma$  answers "how far apart are typical *observations*".

$\sigma$  and SE are different quantities. More data does nothing to  $\sigma$  — the world stays as variable as it was — and shrinks SE.

# Precision Costs Data

| Multiply the data by... | ...and the standard error is multiplied by |
|-------------------------|--------------------------------------------|
| 2                       | 0.71                                       |
| <b>4</b>                | <b><math>\frac{1}{2}</math></b>            |
| 25                      | $\frac{1}{5}$                              |
| <b>100</b>              | <b><math>\frac{1}{10}</math></b>           |

The  $\sqrt{n}$  is why the last digit of an estimate is so expensive: halving the error costs four times the data, and a tenfold improvement costs a hundredfold.

# Two Theorems We Have Already Seen Happen

## Law of large numbers

$\bar{X}_n \rightarrow \mu$  as  $n \rightarrow \infty$ .

The estimate converges on the truth — the pile keeps narrowing.

## Central limit theorem

$\bar{X}_n$  is approximately  $N(\mu, \sigma^2/n)$  for large  $n$ .

And we know the *shape* of what is left.

The first says the error goes away; the second says how much of it is left after  $n$  observations. Only the second lets us put a number on our ignorance.

# From a Number to an Interval

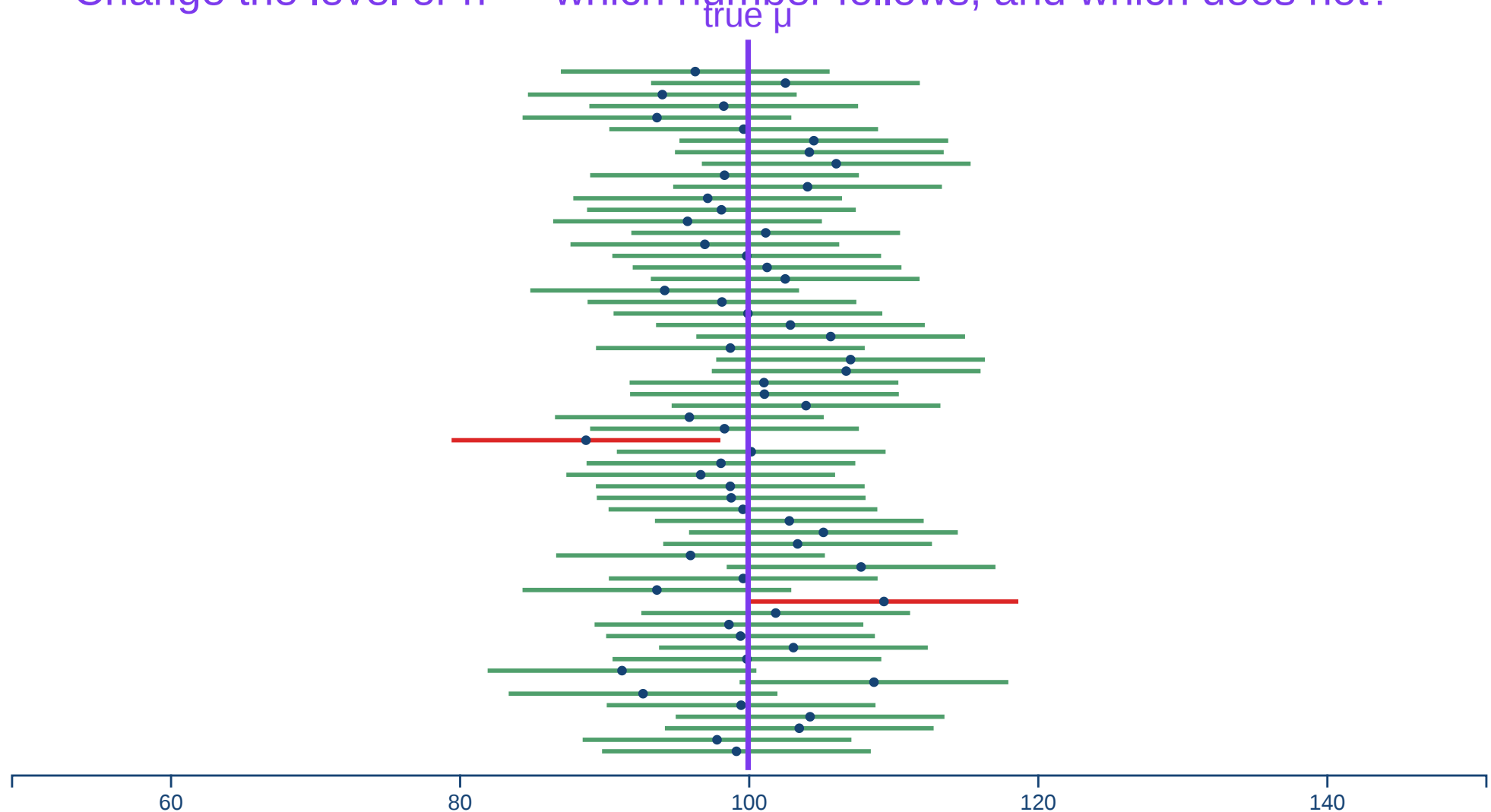
If  $\bar{X} \approx N(\mu, \sigma^2/n)$ , then for 95% of samples  $\bar{X}$  lands within 1.96 SE of  $\mu$  — and therefore  $\mu$  lands within 1.96 SE of  $\bar{X}$ .

$$\bar{x} \pm 1.96 \cdot \frac{\sigma}{n}$$

the estimate — our best single guess  
how confident we insist on being  
how messy the estimate is

# What Does "95% Confident" Mean?

Change the level or n — which number follows, and which does not?



58 of 60 intervals cover  $\mu$  (97%) — nominal 95%

# The Sentence Almost Everyone Gets Wrong

## WHAT IT DOES NOT MEAN

"There is a 95% probability that  $\mu$  lies in **this** interval."

$\mu$  is a fixed number. It is either in this interval or it is not — there is no probability left to have.

## WHAT IT MEANS

"95% of the intervals produced **by this procedure** contain  $\mu$ ."

The probability is a property of the recipe, evaluated before the data is seen.

# Where This Leaves Us

The introduction distinguished two kinds of uncertainty. We can now measure one of them.

- $\sigma$  — how variable the world is. **Aleatoric**. Unchanged by data.
- $SE = \sigma / \sqrt{n}$  — how badly we know our own estimate.  
**Epistemic**. Shrinks with data.

"More data helps" was a slogan.

It is now a rate:  $1/\sqrt{n}$ .



# Mini-Exercise

A shop measures the time customers spend at a counter. From  $n = 100$  customers:  $\bar{x} = 180$  s, and past records give  $\sigma = 60$  s.

1. What is the standard error of the mean?
2. Give an approximate 95% confidence interval.
3. The manager wants the interval to be **half** as wide. How many customers?

# Mini-Exercise

A shop measures the time customers spend at a counter. From  $n = 100$  customers:  $\bar{x} = 180$  s, and past records give  $\sigma = 60$  s.

1. What is the standard error of the mean?
2. Give an approximate 95% confidence interval.
3. The manager wants the interval to be **half** as wide. How many customers?

1.  $SE = 60/\sqrt{100} = 6$  s

2.  $180 \pm 1.96 \cdot 6 \approx [168, 192]$  s

3. Half the width needs twice the  $\sqrt{n}$ , so **four times the data**:  $n = 400$ .

Note what did *not* change in part 3:  $\sigma = 60$  s. Customers stay as variable as they were — only our knowledge of their average improves.

# Takeaway

A statistic is a **random variable**. Its distribution over repeated samples is the sampling distribution; its spread is the **standard error**,  $\sigma/\sqrt{n}$ . A confidence interval reports that spread — and its percentage describes the **procedure**, not the one interval in front of you.